

Network Protocols Assignment

Task

Fill the table below with information about **top 100 network protocols**.

Protocol	Layer	Port Number	TCP/UDP	Encryption	Usage
HTTP	Application	80	TCP	No	Web browsing
HTTPS	Application	443	TCP	Yes (TLS/SSL)	Secure web browsing
FTP	Application	21 (control), 20 (data)	TCP	No	File transfer
FTPS	Application	21, 990 (Implicit)	TCP	Yes (TLS/SSL)	Secure file transfer
SFTP	Application	22	TCP	Yes (SSH)	Secure file transfer over SSH
SSH	Application	22	TCP	Yes	Secure remote login, file transfer, tunneling
Telnet	Application	23	TCP	No	Remote terminal access
SMTP	Application	25, 465 (Implicit SSL), 587 (Explicit TLS)	TCP	Optional (STARTTLS/SSL)	Email sending
POP3	Application	110, 995 (SSL)	TCP	Optional (STARTTLS/SSL)	Email retrieval
IMAP	Application	143, 993 (SSL)	TCP	Optional (STARTTLS/SSL)	Email retrieval and synchronization
DNS	Application	53	UDP & TCP	No (unless DoT/DoH)	Domain name resolution
DHCP	Application	67 (server), 68 (client)	UDP	No	Automatic IP address assignment
TFTP	Application	69	UDP	No	Simple file transfer (e.g., booting)
NTP	Application	123	UDP	No	Network time synchronization
SNMP	Application	161 (polling), 162 (trap)	UDP	v1/v2: No, v3: Yes	Network management
LDAP	Application	389, 636 (LDAPS)	TCP	Optional (STARTTLS/SSL)	Directory access
LDAPS	Application	636	TCP	Yes (SSL/TLS)	Secure LDAP
RDP	Application	3389	TCP & UDP	Yes (typically TLS/SSL)	Remote desktop connection
SMB	Application	445	TCP	v3: Yes, Older: Optional/None	Network file and printer sharing
NFS	Application	2049, dynamic via portmapper (111)	TCP & UDP	Optional	Network file system
MQTT	Application	1883, 8883 (SSL)	TCP	Optional (SSL/TLS)	Messaging protocol for IoT
SIP	Application	5060 (non-secure), 5061 (TLS)	UDP & TCP	Optional (TLS)	Session initiation (VoIP, video calls)
H.323	Application	1719, 1720 (H.225)	UDP & TCP	Optional	VoIP signaling and communication
RTP	Application	Typically dynamic (even numbered)	UDP	No (unless SRTP)	Real-time audio and video delivery
SRTP	Application	Typically dynamic	UDP	Yes	Secure RTP
RTCP	Application	Associated with RTP (RTP port + 1)	UDP	No (unless SRTCP)	Control protocol for RTP sessions
IPsec (AH, ESP)	Network	Uses protocols 51 and 50, not ports. IKE on 500 (UDP).	Not a port (Network Layer)	Yes (ESP)	Encrypts and authenticates IP packets
TCP	Transport	All ports used with it	Not a port (Transport Layer)	No	Reliable, connection-oriented data transfer
UDP	Transport	All ports used with it	Not a port (Transport Layer)	No	Unreliable, connectionless data transfer
ARP	Data Link (layer 2.5)	N/A	N/A	No	Maps IP addresses to MAC addresses
ICMP	Network	N/A	N/A	No	Control and error reporting messages
TLS	Transport / Security	443, dynamic	TCP	Yes	Secures connection between two endpoints
SSH-FTP	Application	22	TCP	Yes (SSH)	Another name for SFTP
BGP	Application	179	TCP	Optional (MD5/TSIG)	Dynamic routing between Autonomous Systems
OSPF	Network	N/A	Protocol 89	No	Dynamic link-state routing
RIP	Application	520 (RIPv1, v2), 521 (RIPng)	UDP	Optional (v2/ng)	Dynamic distance-vector routing
GTP-C	Application	2123	UDP	Optional (via IPsec)	Control plane for GPRS Tunneling Protocol
GTP-U	Application	2152	UDP	Optional (via IPsec)	User plane for GPRS Tunneling Protocol
Kerberos	Application	88	UDP & TCP	Yes	Network authentication
RADIUS	Application	1812 (auth), 1813 (acct)	UDP	Optional (v. RADIUS over TLS)	Authentication, Authorization, Accounting
Diameter	Application	3868	TCP & SCTP	Optional (TLS/DTLS)	Evolution of RADIUS AAA protocol
WHOIS	Application	43	TCP	No	Lookup information on domains
NetBIOS	Application	137, 138, 139	UDP & TCP	No	Legacy network communication
L2TP	Data Link / Security	1701	UDP (via IPsec)	No (typically paired with IPsec)	Tunneling protocol (VPNs)

PPTP	Data Link / Security	1723	TCP (for control)	N/A (requires GRE - protocol 47)	Legacy tunneling protocol
GRE	Network	N/A	Protocol 47	No	Encapsulates various network layer protocols
BOOTP	Application	67 (server), 68 (client)	UDP	No	Predecessor to DHCP
AFP	Application	548	TCP	No (legacy)	Apple Filing Protocol
COAP	Application	5683, 5684 (DTLS)	UDP	Optional (DTLS)	Lightweight M2M protocol
AMQP	Application	5672, 5671 (SSL)	TCP	Optional (SSL/TLS)	Message-oriented middleware protocol
STUN	Session	3478, 5349 (TLS)	UDP & TCP	Optional (TLS)	Discovers public IP behind NAT
TURN	Application	3478, 5349 (TLS)	UDP & TCP	Optional (TLS)	Relays traffic for STUN/NAT traversal
ICE	Application	Dynamic	UDP	No	Framework for finding connection paths
QUIC	Transport / Security	443	UDP	Yes (TLS 1.3)	Built-in encryption and stream multiplexing
HTTP/3	Application	443	UDP (uses QUIC)	Yes (TLS 1.3 via QUIC)	Next version of HTTP over QUIC
DoT	Application	853	TCP	Yes (TLS)	DNS over TLS
DoH	Application	443	TCP	Yes (HTTPS)	DNS over HTTPS
NNTP	Application	119, 563 (NNTPS)	TCP	Optional (STARTTLS/SSL)	Network News Transfer Protocol
NNTPS	Application	563	TCP	Yes (SSL/TLS)	Secure NNTP
SRTCP	Application	associated with SRTP (SRTP port + 1)	UDP	Yes	Secure RTCP
SLP	Application	427	UDP & TCP	No	Service Location Protocol
MSNP	Application	1863	TCP	No	Microsoft Notification Protocol (MSN Messenger)
XMPP (client)	Application	5222	TCP	Optional (STARTTLS/SSL)	Extensible Messaging and Presence Protocol
XMPP (server)	Application	5269	TCP	Optional (STARTTLS/SSL)	Inter-server XMPP communication
Syslog	Application	514 (UDP), 1468 (TCP)	UDP & TCP	Optional (TLS)	Network logging
Syslog-TLS	Application	6514	TCP	Yes (TLS)	Secure Syslog
TACACS+	Application	49	TCP	Yes (encrypts entire packet)	Legacy AAA protocol
IPX	Network	N/A	Protocol 17	No	Legacy Novell internetwork protocol
SPX	Transport	N/A	N/A (part of IPX stack)	No	Legacy sequenced packet exchange
AppleTalk	Network / Data Link	N/A	N/A	No	Legacy Apple suite
H.248	Application	2944	UDP & TCP	Optional	Media Gateway Control Protocol
MGRP	Network	Protocol 126	Protocol 126	No	Mobile Group Resolution Protocol
DCCP	Transport	N/A (Transport Layer)	N/A (Transport Layer)	No	Datagram Congestion Control Protocol
SCTP	Transport	N/A (Transport Layer)	N/A (Transport Layer)	Optional (DTLS)	Stream Control Transmission Protocol
DTLS	Transport / Security	Dynamic	UDP	Yes	Secures connection between two endpoints
RDM	Application	Dynamic	UDP	No	Remote Device Management for lighting
sFlow	Application	6343	UDP	No	Sampling protocol for network statistics
NetFlow	Application	2055, dynamic	UDP	No	IP flow statistics gathering
DDS	Application	Dynamic	UDP	Optional (v. DDS-Security)	Data Distribution Service for Real-Time
IGMP	Network	N/A	Protocol 2	No	Management of multicast group memberships
MPLS	Network / Data Link (Layer 2.5)	N/A	N/A	No	Multi-Protocol Label Switching
LDP	Network / Application	646	UDP & TCP	No	Label Distribution Protocol (for MPLS)
IP Version 4	Network	N/A	Protocols 4, 1, 6, 17, etc.	No	Foundational internet addressing
IP Version 6	Network	N/A	N/A	No	Foundational internet addressing (v6)
LISP	Network / Application	4341 (data), 4342 (control)	UDP	Optional	Locator/ID Separation Protocol
VXLAN	Network	4789	UDP	Optional	Network virtualization tunneling
NVGRE	Network	Protocol 47	GRE	No	Network virtualization using GRE
Geneve	Network	6081	UDP	Optional	Generic Network Virtualization Encapsulation
LACP	Data Link	N/A (Link Layer)	Protocol 802.3ad	No	Link Aggregation Control Protocol
LLDP	Data Link	N/A (Link Layer)	EtherType 0x88CC	No	Link Layer Discovery Protocol
CDP	Data Link	N/A (Link Layer)	EtherType 0x2000	No	Cisco Discovery Protocol
STP	Data Link	N/A (Link Layer)	EtherType 0x01	No	Spanning Tree Protocol (IEEE 802.1D)
VLAN tagging	Data Link	N/A (Link Layer)	EtherType 0x8100 (802.1Q)	No	Creates virtual LANs
Wi-Fi	Data Link / Physical	N/A (Link Layer)	EtherType dynamic	No (unless layer 3/2 crypto like WPA)	Wireless network communication (IEEE 802.11)
Bluetooth	Data Link / Physical	N/A (Link Layer)	Protocol	Yes (Bluetooth pairing)	Short-range wireless

			dynamic		
NFC	Data Link / Physical	N/A (Link Layer)	N/A	No	Near-field communication
CAN bus	Data Link / Physical	N/A (Controller Layer)	N/A	No	Controller Area Network (automotive)
PCI Express	Data Link / Physical	N/A (Bus Layer)	N/A	No	Peripheral component interconnect
SAS	Data Link / Physical	N/A (Storage Layer)	N/A	No	Serial Attached SCSI (storage)
SATA	Data Link / Physical	N/A (Storage Layer)	N/A	No	Serial Advanced Technology Attachment (storage)

Instructions

- Fill all columns
- Keep answers short
- Write correct port numbers and layers
- Submit in Word or PDF